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Assignment 6A

Aim

Write a program to imlement insert, delete and taversal on a singly linked list

Theory

A linked list is an ordered collection of data where each element (generally called nodes) contains the location of the next element in the list.

Each node has two parts - ~~one~~ the data part and the link which connects it to the next one which is usually called next

Insertion.

In a linked list mean adding a new node to the list A new node is created, its data is stored and its address is connected to the next node. The previous node's link is updated to point to this new node.

Deletion

Means removing a node from the linked list. The link of the previous node is changed so that it skips the node to be deleted. Then the memory of the deleted node is freed.

Traversal

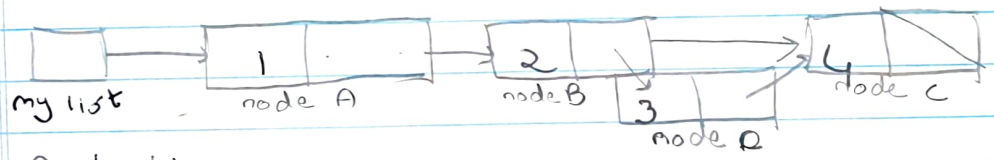
It means visiting each node of the linked list

one by one. We start from the first node (head) and follow the links using help pointer until we reach NULL. This is usually done to display or search elements in the list.

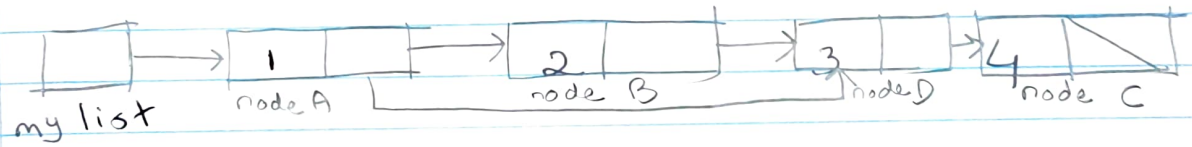
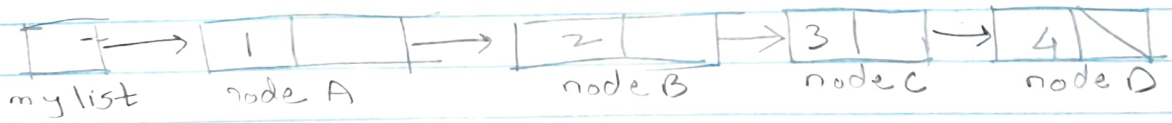
Time Complexity
Insertion & Deletion - O(1)
Traversal - O(n)

Conclusion
The concept of linked lists is applied to evaluate problems in a variety of applications

e.g. Insertion



Deletion



Traversal